

Physical Chemistry (I)

Lecture 08

Thermochemistry

Second Law of Thermodynamics



In the Last Class

- Enthalpy
 - Definition
 - Differential form
 - Adiabatic changes
- Thermochemistry
 - Total differentials
 - Response functions

Enthalpy H

$$H = U + pV$$

- Energy required to create a system (U) and to make a room for the system (pV)
- Meaning: $dH = \delta q_p$ or $\Delta H = q_p$

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p$$

Last class

Differential Form of H

For $H = H(p, T)$,

$$dH = \left(\frac{\partial H}{\partial p} \right)_T dp + \left(\frac{\partial H}{\partial T} \right)_p dT$$

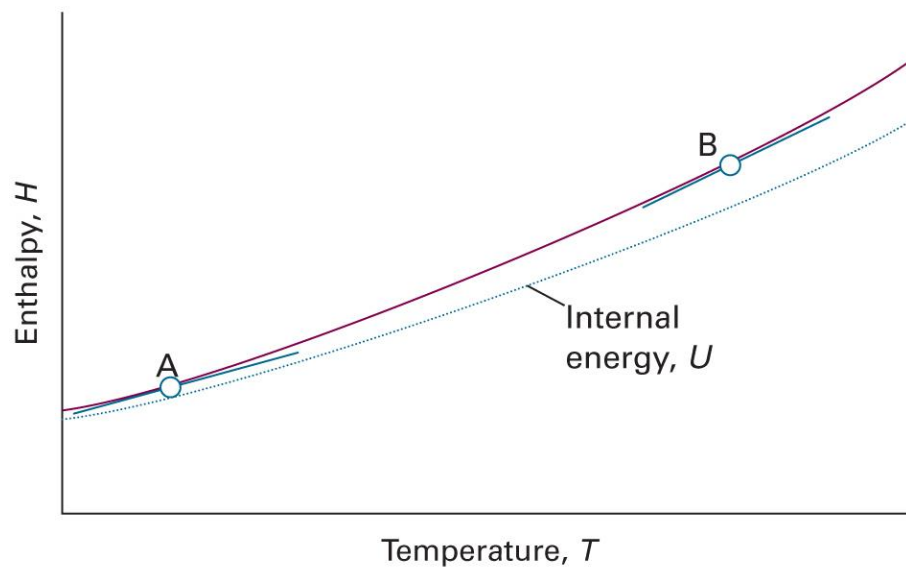
$$dH = -C_p \mu dp + C_p dT$$

(Compare: $dU = \pi_T dV + C_V dT$)

Last class

C_V and C_p

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V, \quad C_p = \left(\frac{\partial H}{\partial T} \right)_p$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2B.3)

Adiabatic Changes

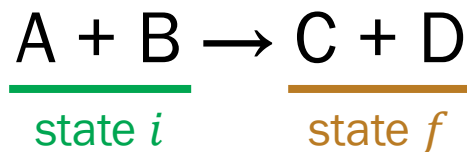
$$\delta q = 0$$

For a perfect gas whose C_V is constant,

- Work: $w = C_V(T_f - T_i)$
- Relations in a reversible change:
 - $VT^c = \text{constant}$ ($c = C_V/nR$)
 - $pV^\gamma = \text{constant}$ ($\gamma = 1 + 1/c$)

Thermochemistry

“The study of the energy transferred as heat during the course of chemical reactions”



$$q_V = \Delta U$$

$$q_p = \Delta H$$

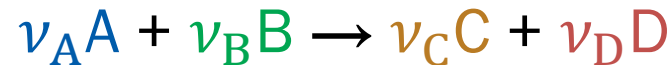
Standard Enthalpy, $H^\ominus$

Enthalpy H of a substance in its **standard state**

- **Standard state** of a substance: its pure form at 1 bar
 - Defined for a specific temperature
 - If temperature is not specified, use the conventional temperature of 298.15 K
- Standard enthalpy change $\Delta H^\ominus = H_f^\ominus - H_i^\ominus$

Last class

Standard Reaction Enthalpy



- Standard reaction enthalpy $\Delta_r H^\ominus$: standard enthalpy change $\Delta H^\ominus$ for the reaction per ...
 - ν_A moles of consumed A
 - ν_B moles of consumed B
 - ν_C moles of produced C
 - ν_D moles of produced D

Last class

Hess's Law

“The standard reaction enthalpy is the sum of the values for the individual reactions into which the overall reaction may be divided.”

This holds because H is a **state function**

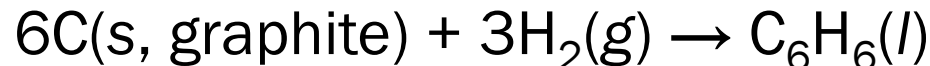


Last class

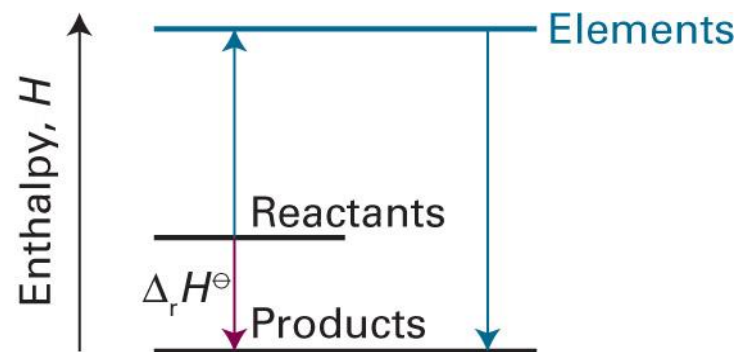
Standard Enthalpy of Formation

- Standard enthalpy of formation $\Delta_f H^\ominus$ of a substance: $\Delta_r H^\ominus$ for the formation of the compound from its elements in their reference states
- Reference state of an element: its most stable state at the specified temperature and 1 bar

e.g. $\Delta_f H^\ominus$ of liquid benzene at 298 K: +49.0 kJ mol⁻¹



$\Delta_f H^\ominus$ and $\Delta_r H^\ominus$



$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) \Delta_f H^\ominus(i) - \sum_{\text{reactants } j} \nu(j) \Delta_f H^\ominus(j)$$

Temperature Dependence

$$dH = C_p dT$$

$$\int_{\text{state } i}^{\text{state } f} dH = \int_{\text{state } i}^{\text{state } f} C_p dT$$

$$H_f - H_i = \int_{T_i}^{T_f} C_p dT$$

$$H(T_f) - H(T_i) = \int_{T_i}^{T_f} C_p dT$$

Temperature Dependence

For $\nu_A A + \nu_B B \rightarrow \nu_C C + \nu_D D$,

$$\Delta_r H^\ominus(T_f) = \nu_C H_m^\ominus(C, T_f) + \nu_D H_m^\ominus(D, T_f) - \nu_A H_m^\ominus(A, T_f) - \nu_B H_m^\ominus(B, T_f)$$

Since $H(T_f) = H(T_i) + \int_{T_i}^{T_f} C_p dT$,

$$H_m^\ominus(T_f) = H_m^\ominus(T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus dT$$

$$\Delta_r H^\ominus(T_f)$$

$$\begin{aligned} &= \nu_C \left\{ H_m^\ominus(C, T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus(C) dT \right\} + \nu_D \left\{ H_m^\ominus(D, T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus(D) dT \right\} \\ &\quad - \nu_A \left\{ H_m^\ominus(A, T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus(A) dT \right\} - \nu_B \left\{ H_m^\ominus(B, T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus(B) dT \right\} \end{aligned}$$

Temperature Dependence

$$\Delta_r H^\ominus(T_f) = \nu_C H_m^\ominus(C, T_i) + \nu_D H_m^\ominus(D, T_i) - \nu_A H_m^\ominus(A, T_i) - \nu_B H_m^\ominus(B, T_i) \\ + \int_{T_i}^{T_f} \underbrace{\{\nu_C C_{p,m}^\ominus(C) + \nu_D C_{p,m}^\ominus(D) - \nu_A C_{p,m}^\ominus(A) - \nu_B C_{p,m}^\ominus(B)\}}_{\Delta_r C_p^\ominus} dT$$

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \int_{T_i}^{T_f} \Delta_r C_p^\ominus dT$$

Kirchhoff's Law

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \int_{T_i}^{T_f} \Delta_r C_p^\ominus dT$$

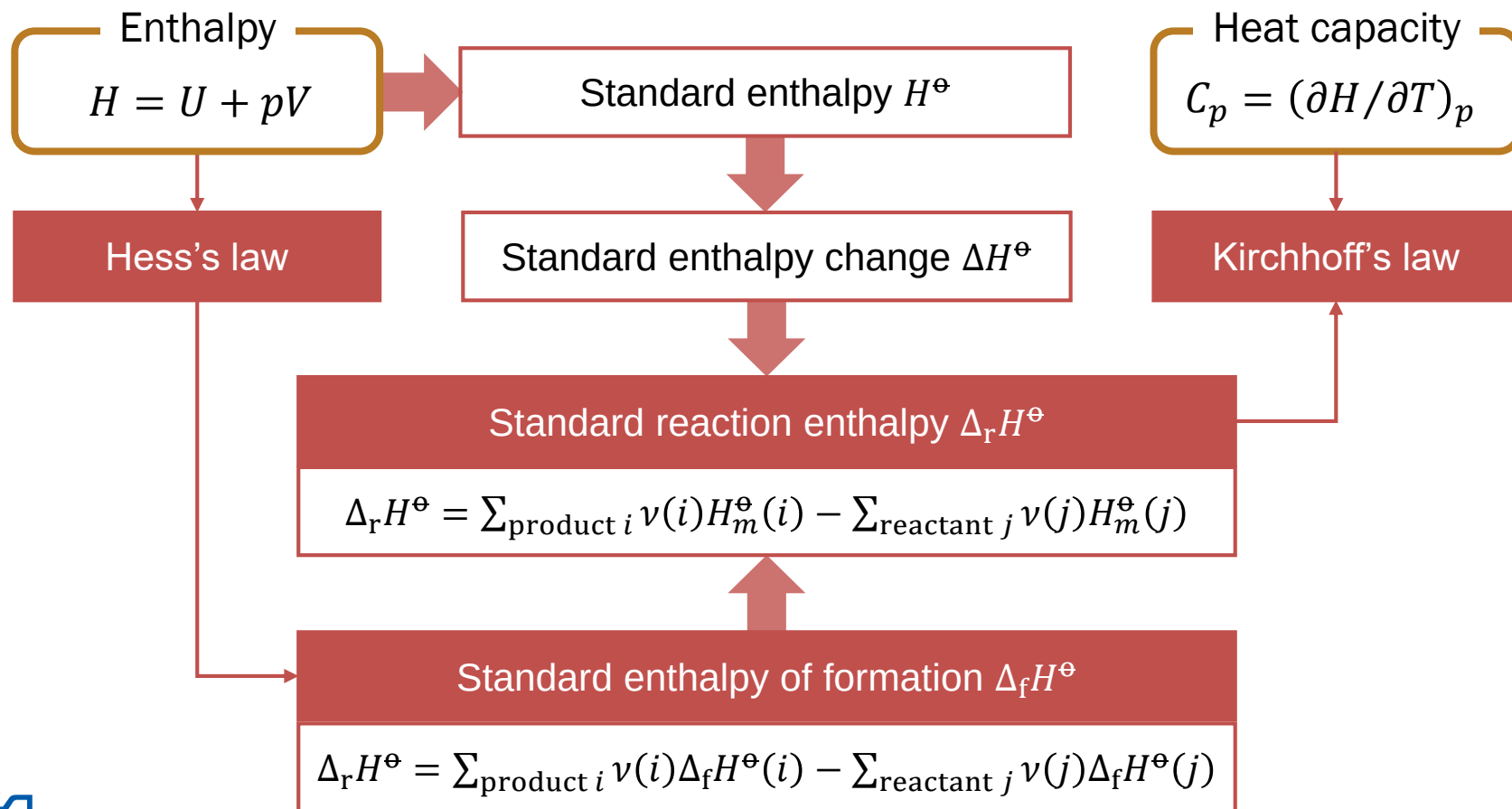
where

$$\Delta_r C_p^\ominus = \sum_{\text{products } i} \nu(i) C_{p,m}^\ominus(i) - \sum_{\text{reactants } j} \nu(j) C_{p,m}^\ominus(j)$$

If $\Delta_r C_p^\ominus$ is constant in $[T_i, T_f]$,

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \Delta_r C_p^\ominus \times (T_f - T_i)$$

Summary for Chapter 3 (3)



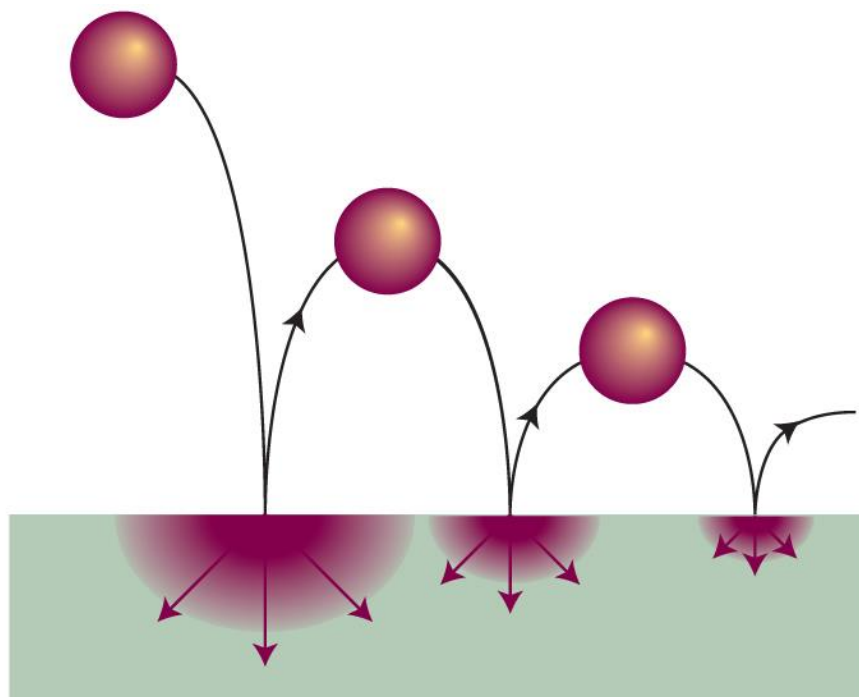
4. Entropy

Chapter Overview

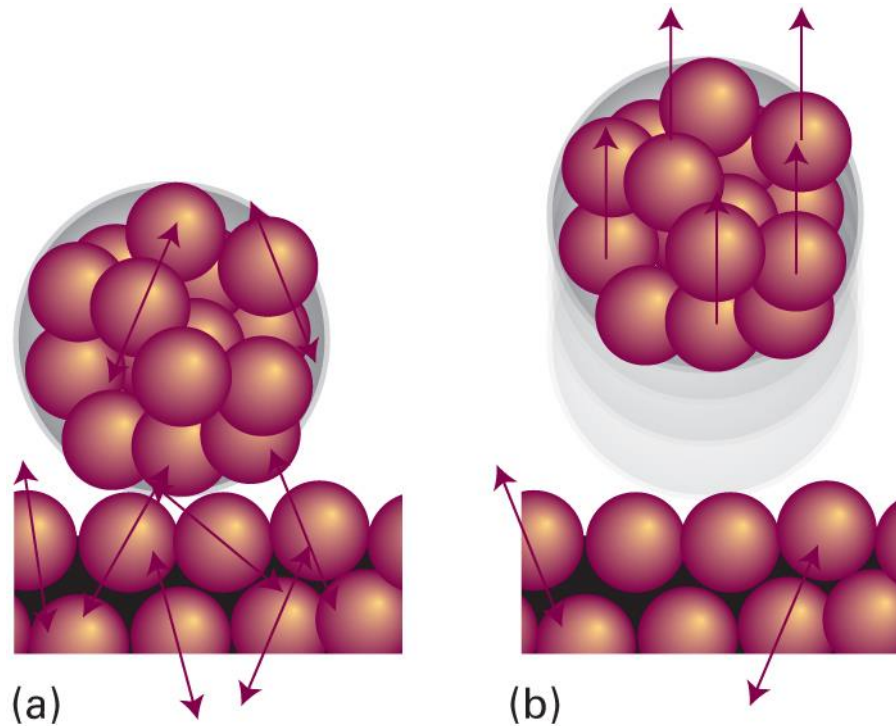
- Second law of thermodynamics
 - Kelvin's statement and Clausius's statement
 - Heat engine and Carnot cycle
- Entropy
 - Definition and properties
 - Third law of thermodynamics
 - Entropy and thermochemistry
- Physical meaning of entropy
 - Thermodynamic meaning
 - Statistical meaning: Boltzmann's formula



“Arrow of Time”

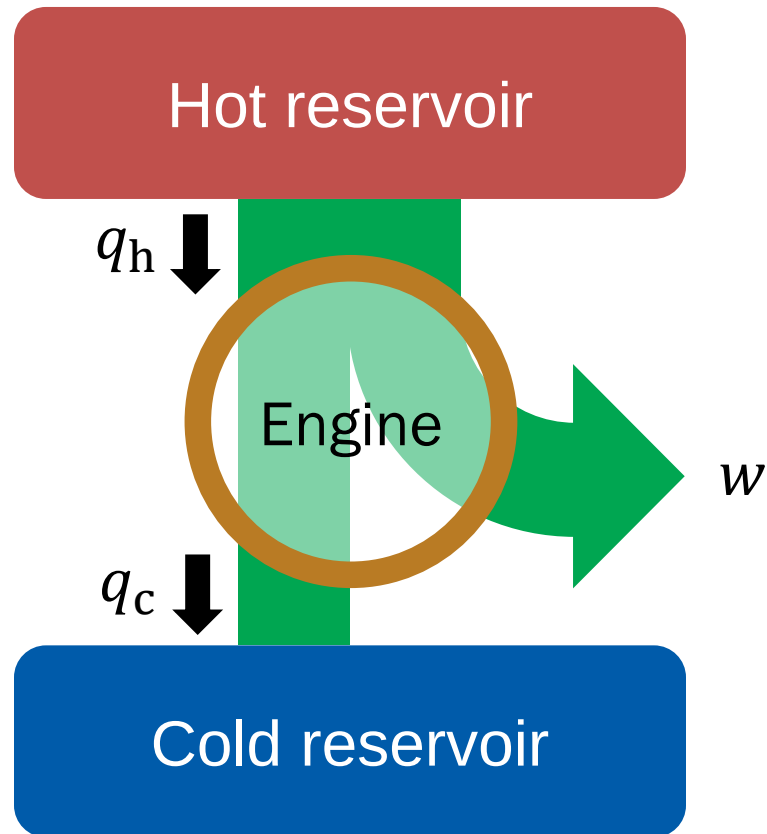


Why?



Heat and work are not equal

Heat Engine

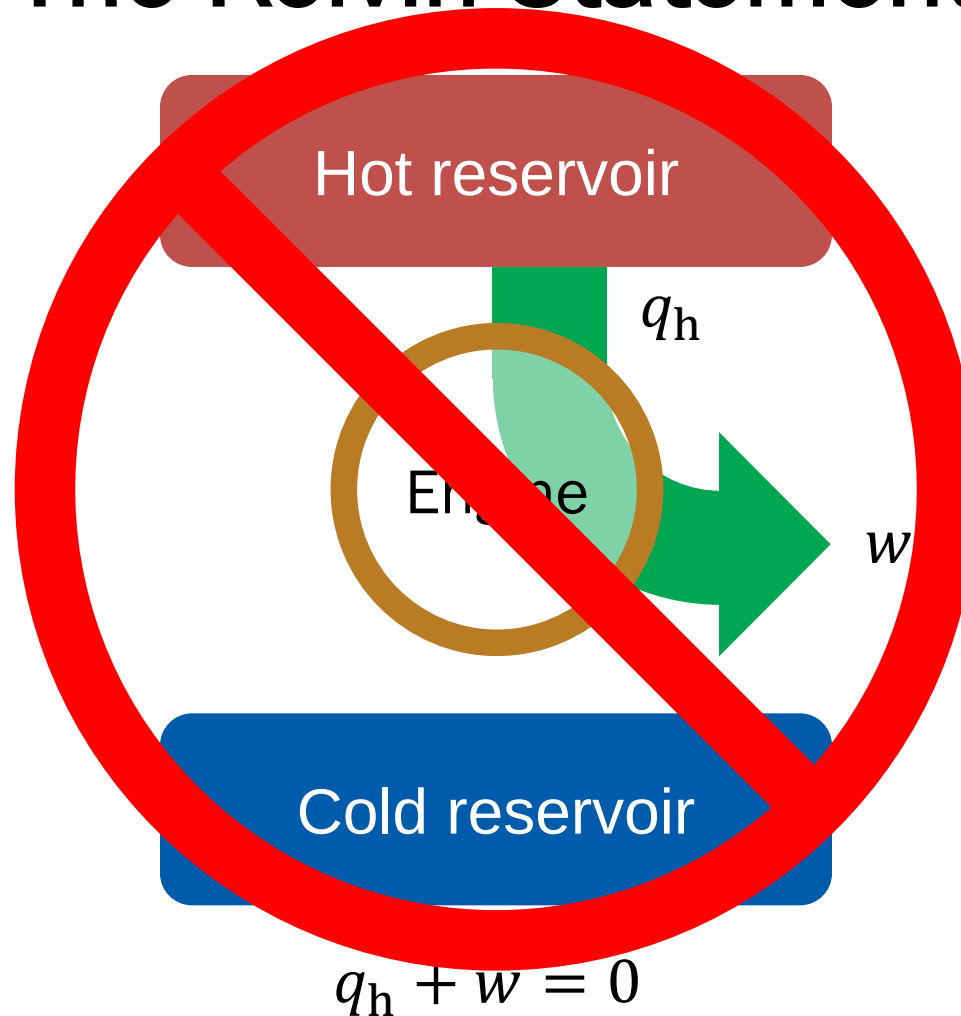


$$q_h + q_c + w = 0$$

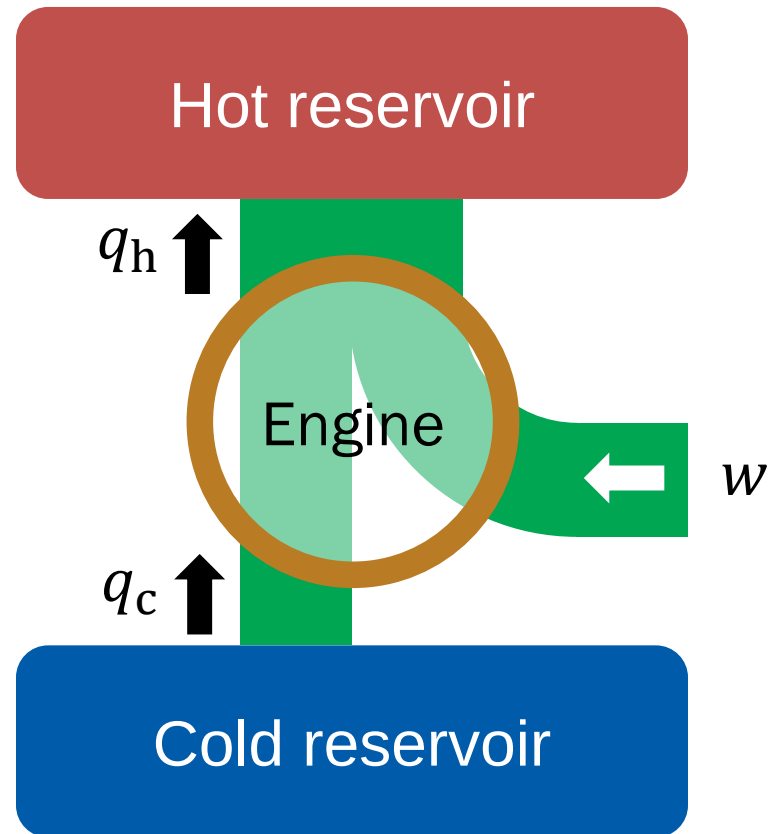
Second Law (Kelvin)

“No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work.”

The Kelvin Statement



“Refrigerator”

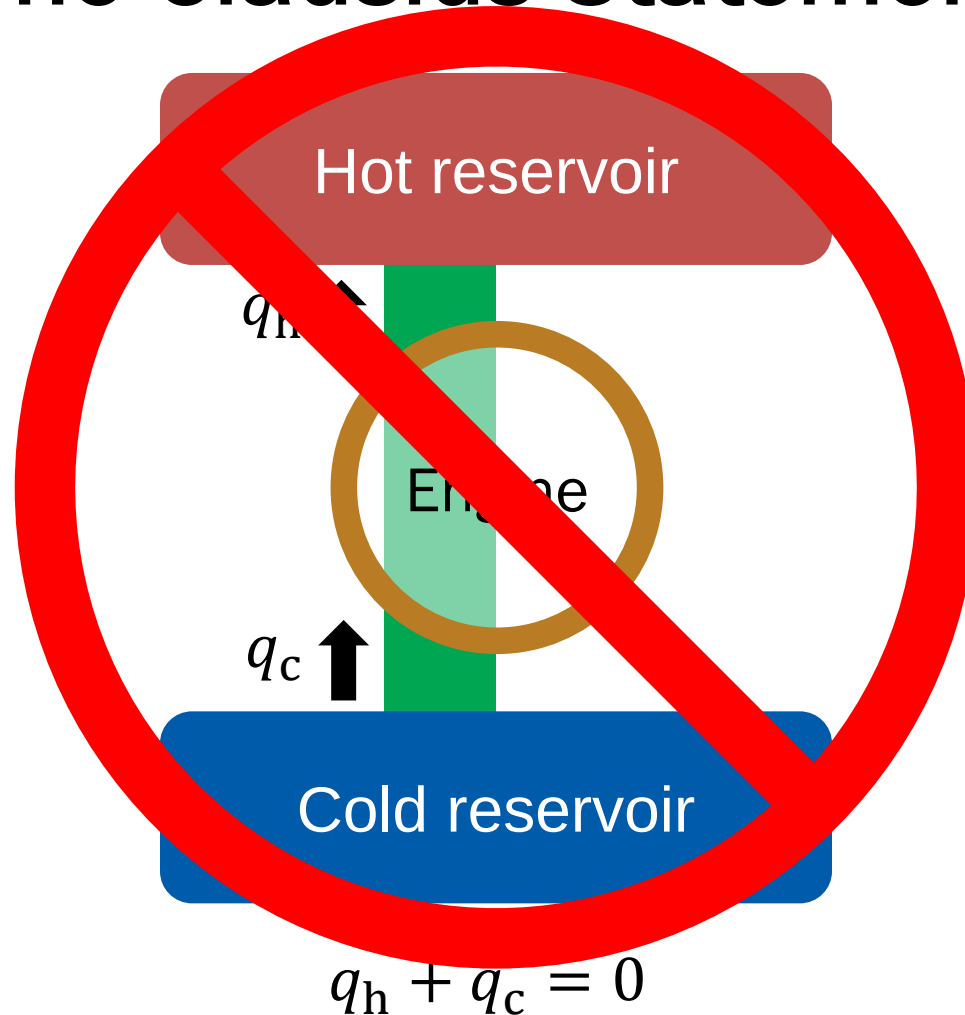


$$q_h + q_c + w = 0$$

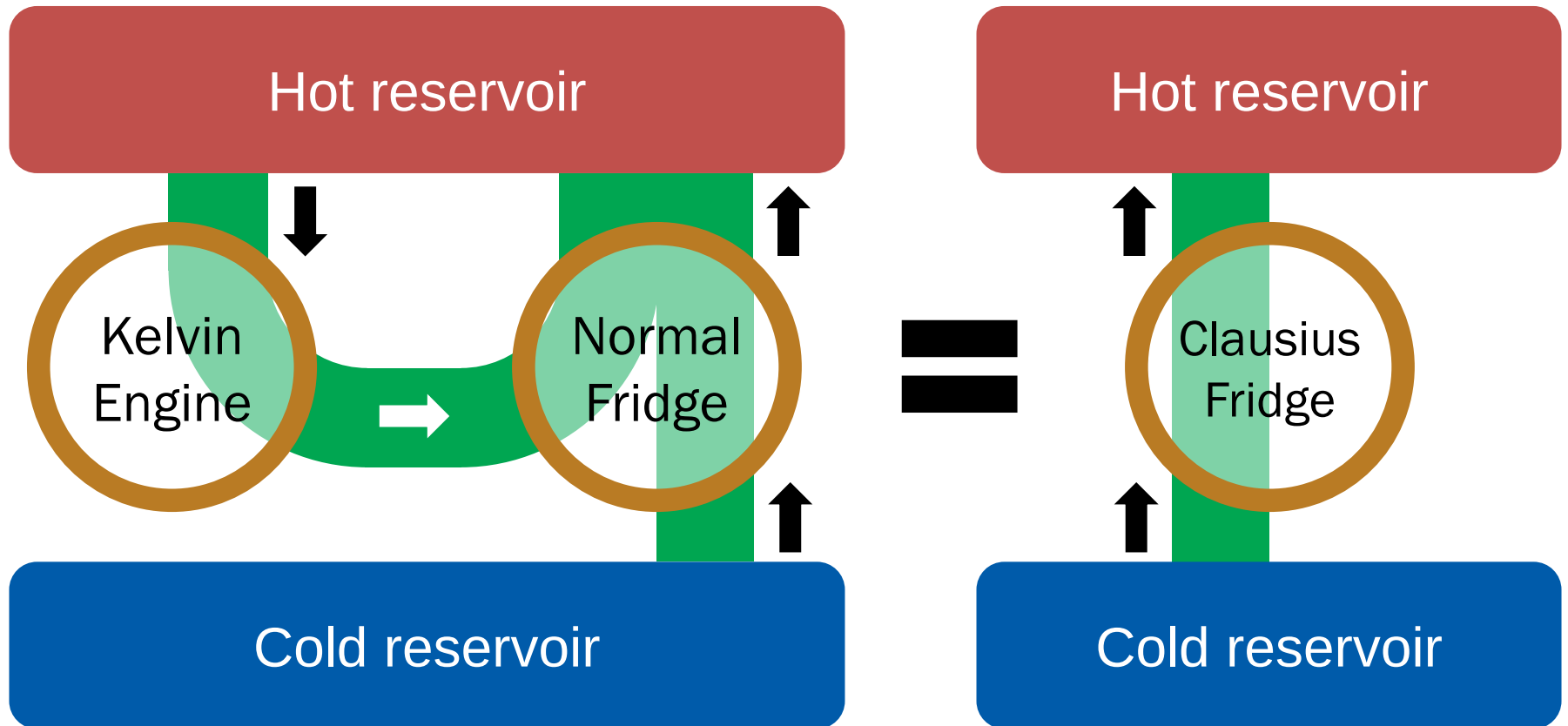
Second Law (Clausius)

“Heat does not flow spontaneously from a cool body to a hotter body.”

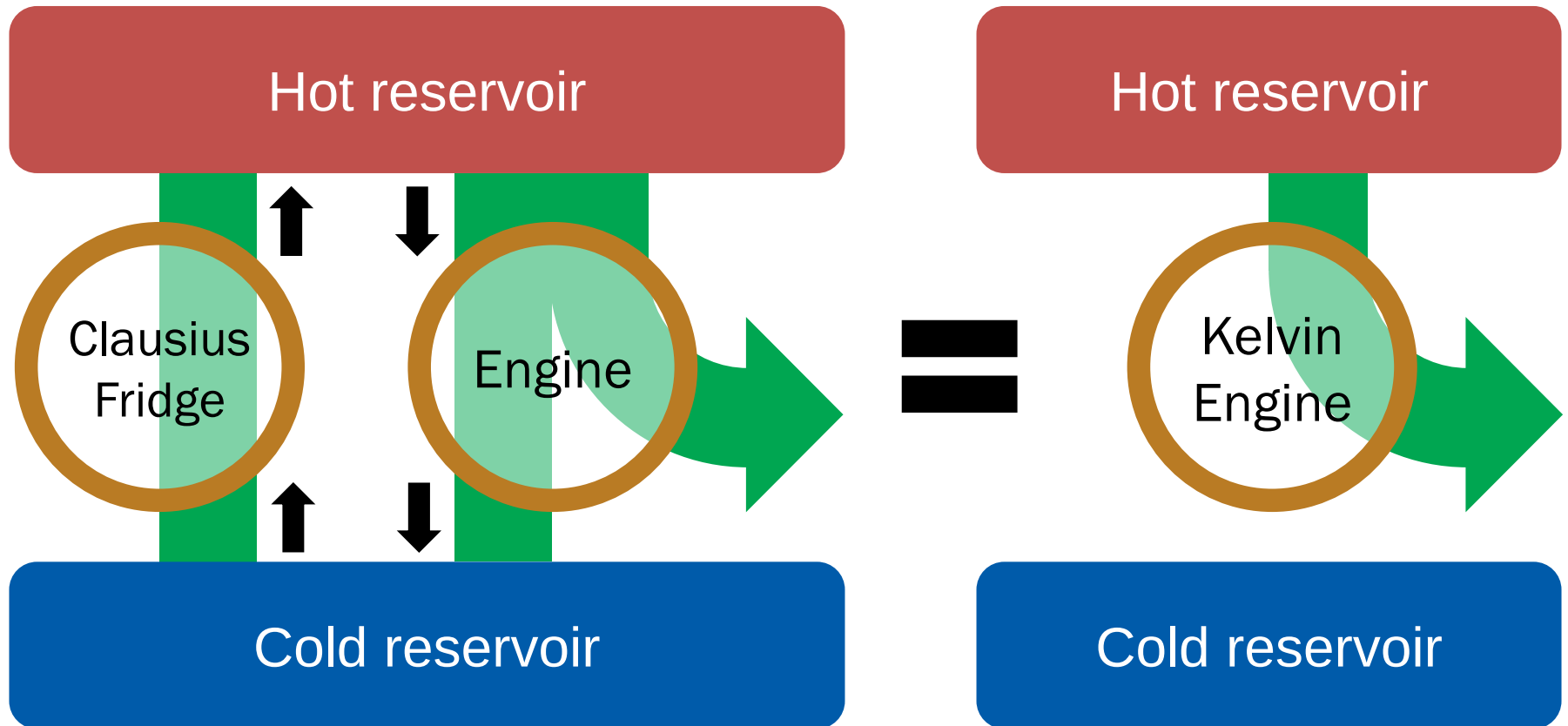
The Clausius Statement



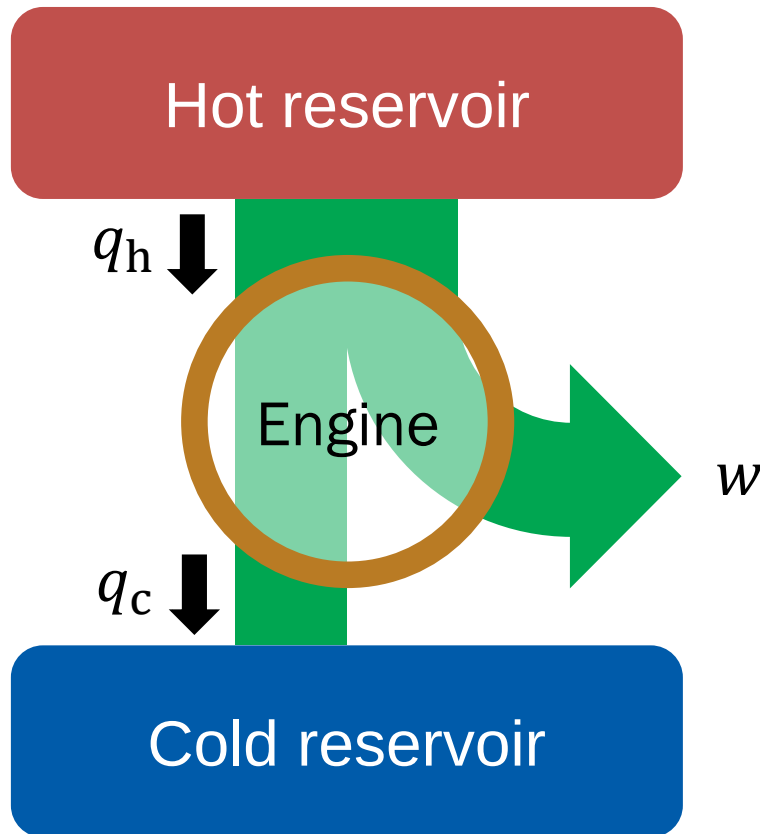
Kelvin $\rightarrow$ Clausius



Clausius $\rightarrow$ Kelvin



Efficiency of the Engine



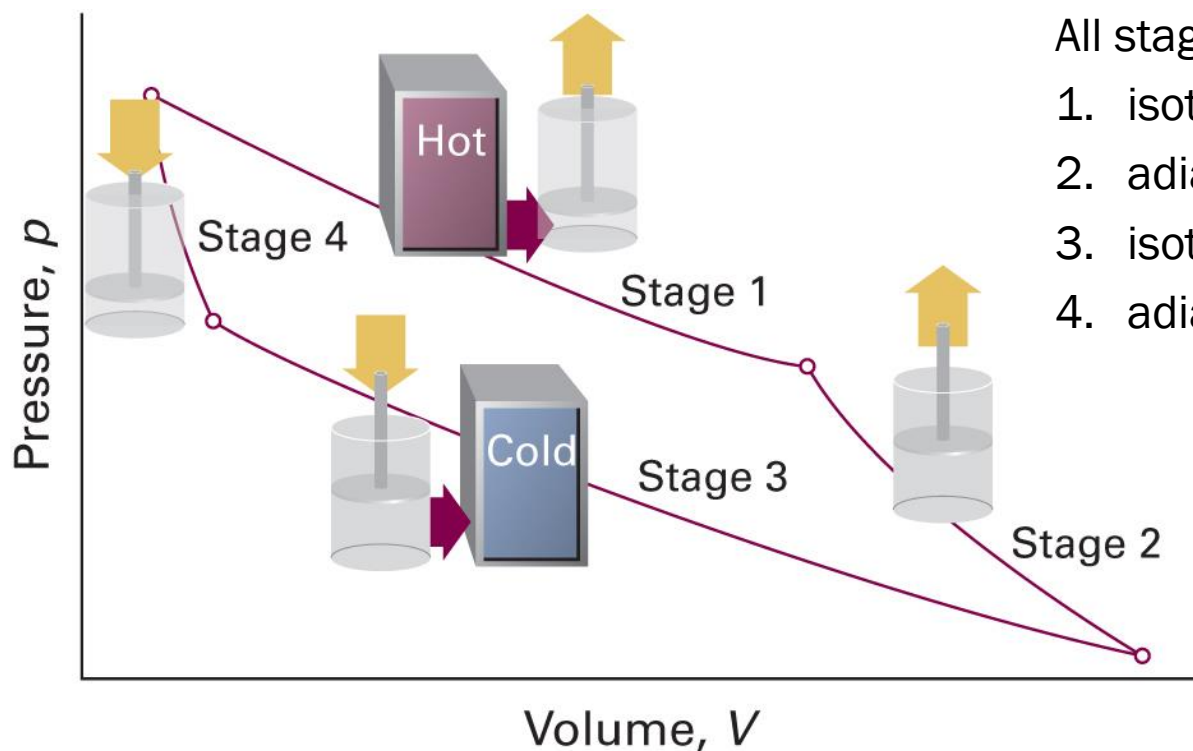
$$\eta = \frac{|w|}{|q_h|}$$

Hence,

$$\eta = \frac{|q_h| - |q_c|}{|q_h|} = 1 - \frac{|q_c|}{|q_h|}$$

$$q_h + q_c + w = 0$$

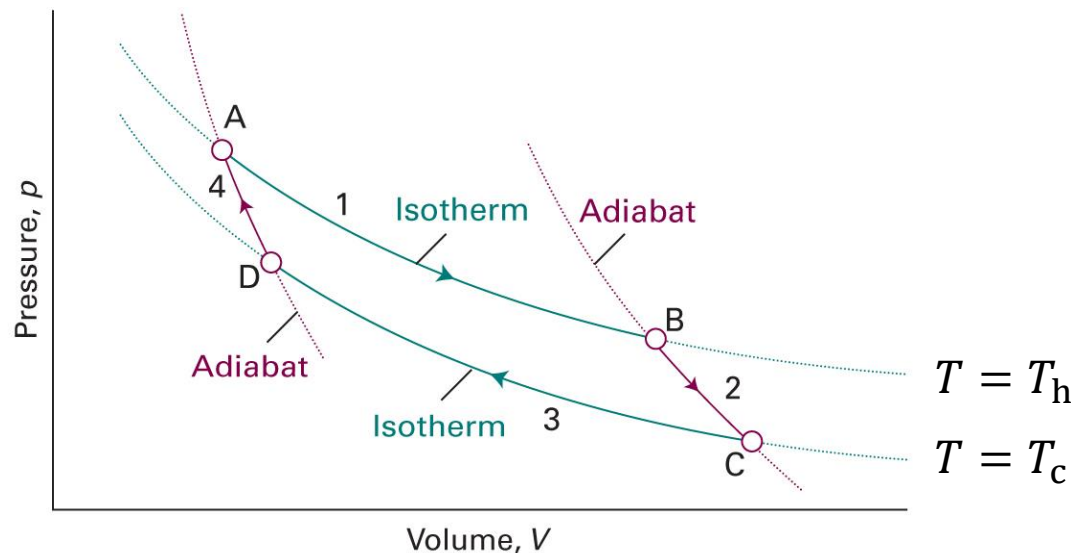
Carnot Cycle



- All stages are reversible;
1. isothermal expansion
 2. adiabatic expansion
 3. isothermal compression
 4. adiabatic compression

Not in Atkins

Perfect Gas in Carnot Engine



1. Isothermal expansion (at T_h)

$$w_1 = -nRT_h \ln(V_B/V_A)$$

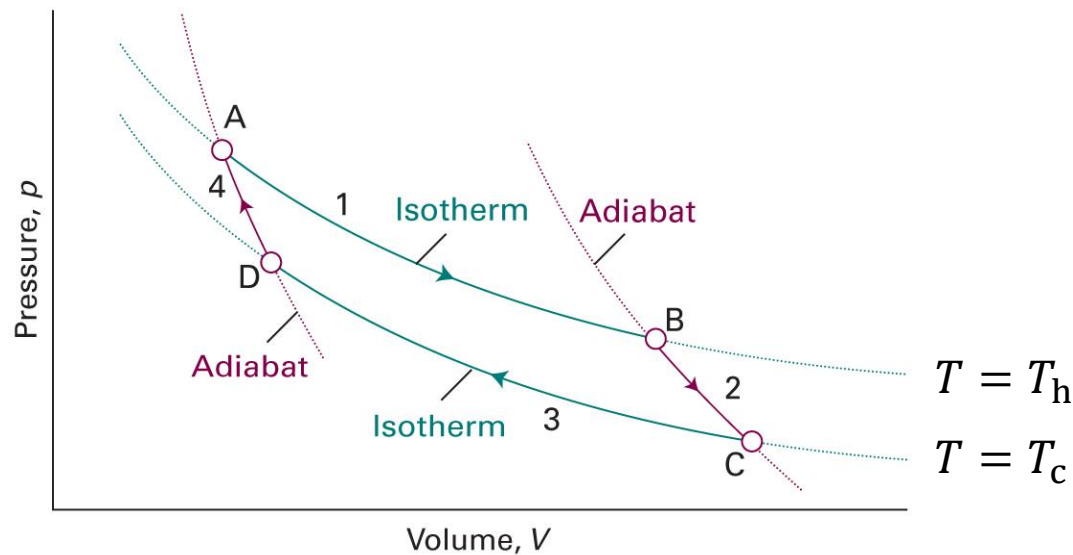
$$\text{Since } \Delta U = 0, q_1 = nRT_h \ln(V_B/V_A)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 3A.9)

Not in Atkins

Perfect Gas in Carnot Engine



2. Adiabatic expansion

$$q_2 = 0$$

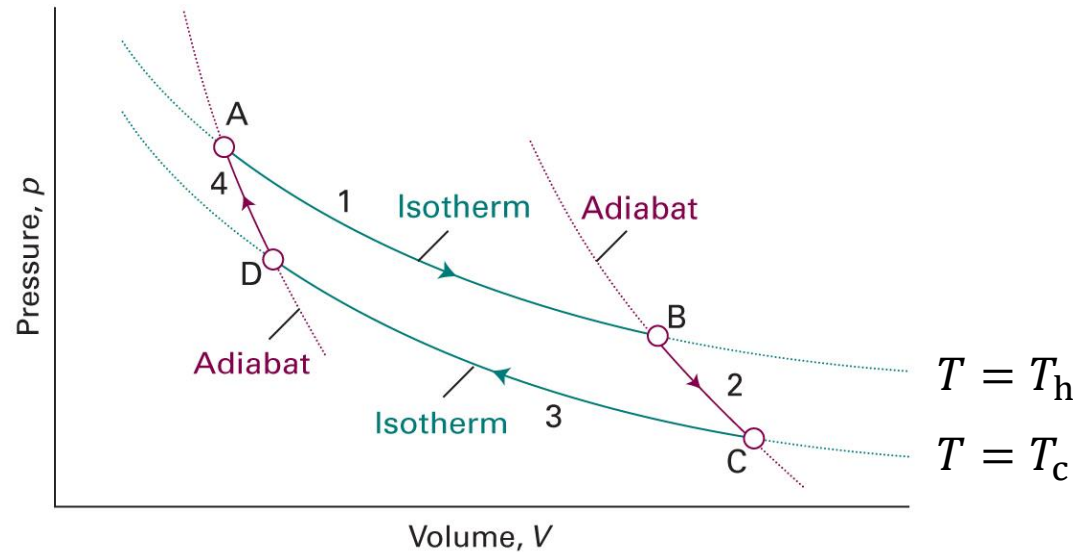
$$w_2 = C_V(T_c - T_h)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 3A.9)

Not in Atkins

Perfect Gas in Carnot Engine



3. Isothermal compression (at T_c)

$$w_3 = -nRT_c \ln(V_D/V_C)$$

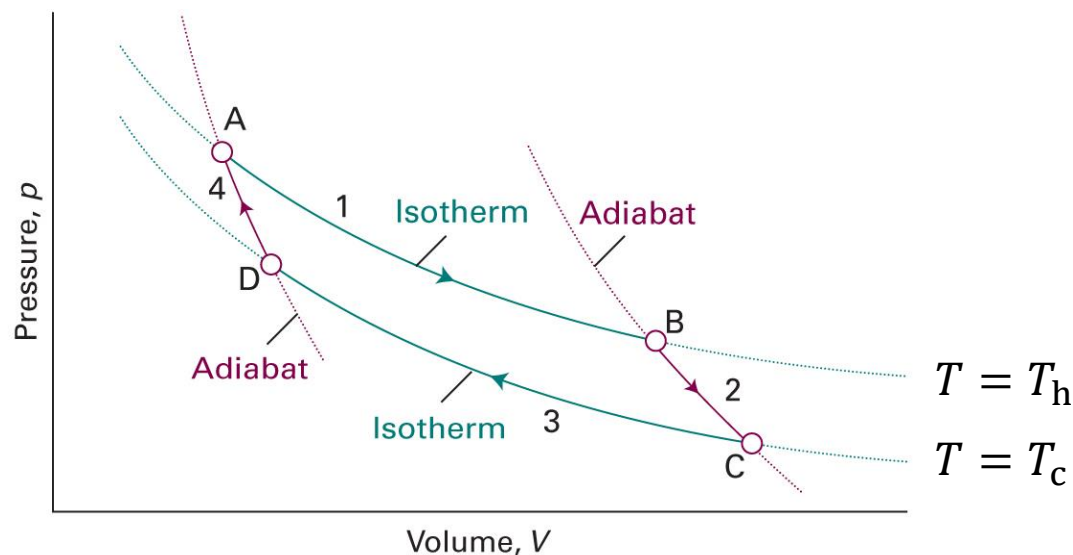
$$\text{Since } \Delta U = 0, q_3 = nRT_c \ln(V_D/V_C)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 3A.9)

Not in Atkins

Perfect Gas in Carnot Engine



4. Adiabatic compression

$$q_4 = 0$$

$$w_4 = C_V(T_h - T_c)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 3A.9)

After One Cycle ...

- Internal energy change $\Delta U = 0$ (state function)
- Enthalpy change $\Delta H = 0$ (state function)

- Heat

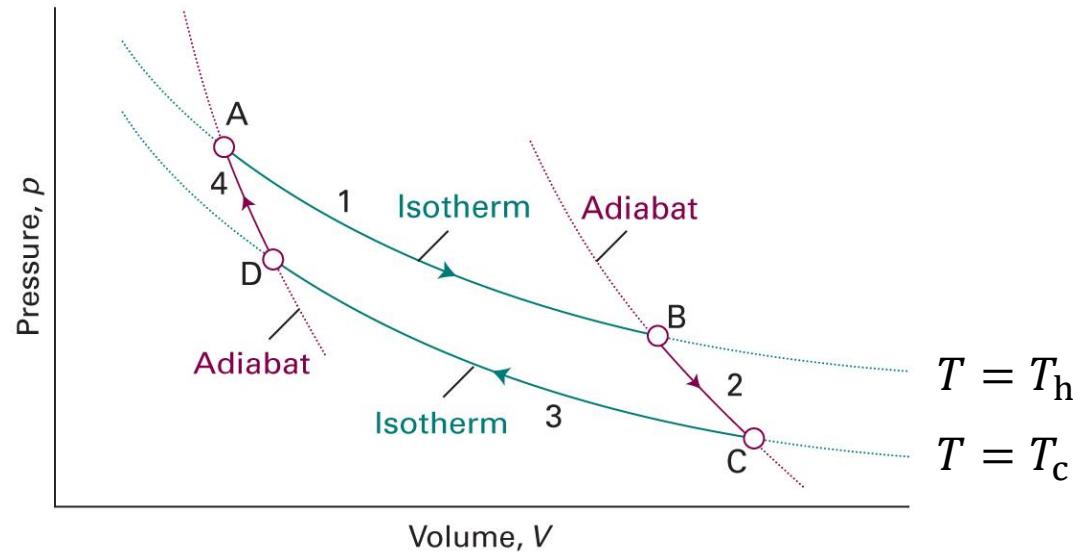
$$q = q_1 + q_2 + q_3 + q_4 = nRT_h \ln(V_B/V_A) + nRT_c \ln(V_D/V_C)$$

- Work

$$w = w_1 + w_2 + w_3 + w_4$$

$$w = -nRT_h \ln(V_B/V_A) - nRT_c \ln(V_D/V_C) = -q$$

Use of Adiabats



2. Adiabatic expansion: $V_B T_h^c = V_C T_c^c$ ($c = C_V/nR$)

4. Adiabatic compression: $V_D T_c^c = V_A T_h^c$

Heat and Work

Since $V_B T_h^c = V_C T_c^c$ and $V_D T_c^c = V_A T_h^c$,

$$\frac{V_B}{V_C} = \left(\frac{T_c}{T_h} \right)^c = \frac{V_A}{V_D}$$

Hence,

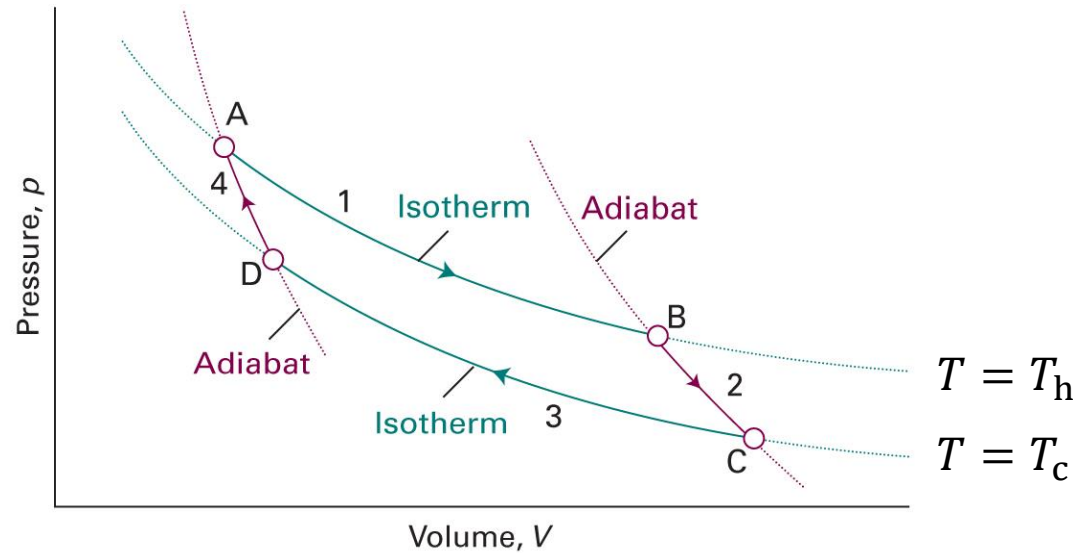
$$q = nRT_h \ln \left(\frac{V_B}{V_A} \right) + nRT_c \ln \left(\frac{V_D}{V_C} \right) = nR(T_h - T_c) \ln \left(\frac{V_B}{V_A} \right)$$

The work is

$$w = -q = -nR(T_h - T_c) \ln \left(\frac{V_B}{V_A} \right)$$

Not in Atkins

Efficiency of Carnot Engine



$$\eta = \frac{|w|}{|q_h|} = \frac{|w|}{|q_1|}$$

Not in Atkins

Efficiency of Carnot Engine

$$\eta = \frac{|w|}{|q_h|} = \frac{|w|}{|q_1|}$$

For $w = -nR(T_h - T_c) \ln(V_B/V_A)$ and $q_1 = nRT_h \ln(V_B/V_A)$,

$$\eta = \frac{nR(T_h - T_c) \ln(V_B/V_A)}{nRT_h \ln(V_B/V_A)} = \frac{T_h - T_c}{T_h} = 1 - \frac{T_c}{T_h}$$

Carnot's Theorem

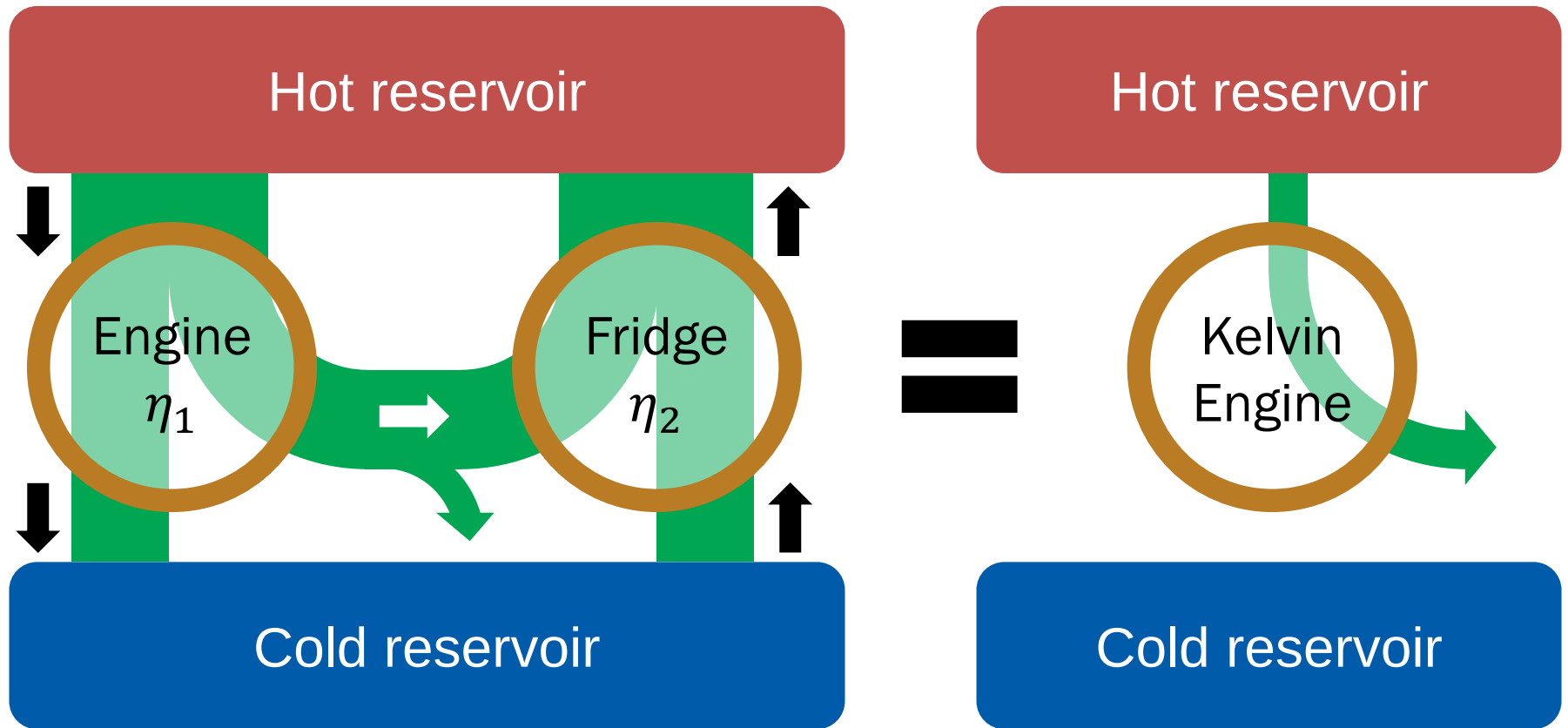
$$\eta = 1 - T_c/T_h \text{ (perfect-gas reversible Carnot engine)}$$

“The efficiency of a reversible Carnot engine depends only on the temperatures of the reservoirs.”

→ independent of the working material

$$\eta = 1 - T_c/T_h \text{ (any reversible Carnot engine)}$$

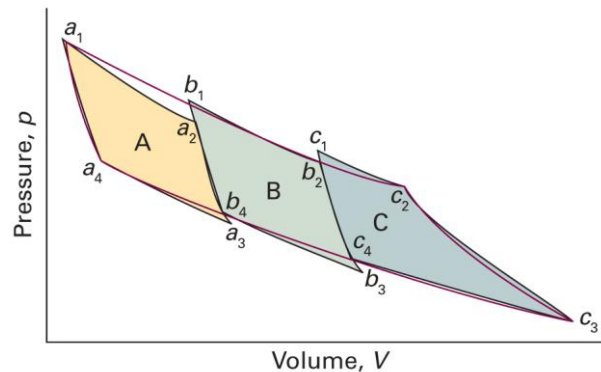
Proof of Equivalence



(if $\eta_1 > \eta_2$)

General Reversible Engine

“All reversible engines have the same efficiency regardless of their construction”



$$\eta = 1 - T_c/T_h \text{ (any reversible engine)}$$

Brief Illustration 3A.3

A certain power station operates with superheated steam at 300 °C ($T_h = 573$ K) and discharges the waste heat into the environment at 20 °C ($T_c = 293$ K). What is its theoretical efficiency?

$$\eta = 1 - \frac{293 \text{ K}}{573 \text{ K}} = 0.489 = 48.9\%$$

One More State Function

$$\eta = 1 - \frac{|q_c|}{|q_h|} = 1 + \frac{q_c}{q_h}$$

For a reversible engine,

$$\eta = 1 - \frac{T_c}{T_h}$$

Hence, for a reversible cycle,

$$\frac{q_h}{T_h} + \frac{q_c}{T_c} = \Delta \left(\frac{q}{T} \right) = 0$$

Entropy

For a reversible cycle,

$$\Delta \left(\frac{q_{\text{rev}}}{T} \right) = 0$$

which implies that q_{rev}/T is a **state function**

Thermodynamic definition of entropy:

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

Entropy Calculation

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

- Entropy change

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

- Molar entropy change $\Delta S_{\text{m}} = \Delta S/n$

Example 3A.1

Calculate the entropy change of a sample of perfect gas when it expands isothermally from a volume V_i to a volume V_f .

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

Since it is an isothermal change,

$$\Delta S = \frac{1}{T} \int_{\text{state } i}^{\text{state } f} \delta q_{\text{rev}} = \frac{q_{\text{rev}}}{T}$$

For an isothermal change of perfect gas,

$$\Delta U = q + w = 0 \text{ and } w = -nRT \ln(V_f/V_i)$$

Example 3A.1

Since $q = nRT \ln(V_f/V_i)$,

$$\Delta S = \frac{q_{\text{rev}}}{T} = nR \ln\left(\frac{V_f}{V_i}\right)$$